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# FaceLock Pro

AI-Powered Privacy & App Locker for macOS

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Complete Technical Report

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**Project Name:** FaceLock Pro (AppLockPro)  
**Version:** 1.0.0  
**Platform:** macOS 14.0+ (Sonoma)  
**Target Hardware:** Apple Silicon (M1, M2, M3, M4)  
**Language:** Swift 5.9+  
**Architecture:** MVVM (Model-View-ViewModel)  
**AI Processing:** 100% Offline

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# Chapter 1

## Introduction

### 1.1 Project Overview

**FaceLock Pro** is a professional macOS desktop application designed to add an intelligent privacy layer to the operating system. By requiring biometric authentication — specifically facial recognition — before allowing access to user-designated applications, it ensures that sensitive data remains secure even if the Mac is left unlocked.

A critical tenet of this project is privacy: the application runs entirely offline. No biometric or personal data is ever uploaded to external servers. It is optimized exclusively for Apple Silicon Macs (M1, M2, M3, M4) to provide fast, efficient, and battery-friendly authentication by leveraging the Apple Neural Engine for on-device machine learning inference.

### 1.2 Problem Statement

macOS provides system-level login security (password, Touch ID, Apple Watch unlock), but once a user is logged in, all applications are freely accessible. This creates a privacy gap in several scenarios:

- A user steps away from an unlocked Mac in a shared workspace.
- A family member or friend borrows the laptop and opens personal applications such as Messages, Photos, or Notes.
- Sensitive applications (banking, email, code editors with proprietary projects) remain unprotected after login.
- Existing third-party solutions often require cloud connectivity, raising privacy and data sovereignty concerns.

FaceLock Pro addresses these gaps by providing application-level biometric protection that operates entirely on-device.

### 1.3 Project Objectives

The core objectives include:

1. **Application Protection:** Allow users to select and protect any installed application.

2. **Facial Recognition:** Unlock applications using real-time AI-powered facial recognition.
3. **Access Prevention:** Prevent unauthorized access to protected applications by intercepting launch events.
4. **Offline Operation:** Ensure 100% offline processing to guarantee user privacy.
5. **Secure Storage:** Store biometric data (face embeddings) encrypted locally using the macOS Keychain.
6. **Fast Authentication:** Provide sub-500-millisecond authentication using Apple Neural Engine hardware acceleration.
7. **Battery Efficiency:** Minimize battery and CPU impact through event-driven architecture.
8. **Native Experience:** Adhere to Apple’s macOS Human Interface Guidelines for a polished user experience.

## 1.4 Scope

### 1.4.1 In Scope

- Native macOS application for Apple Silicon.
- Offline facial recognition using ArcFace (InsightFace) converted to Core ML.
- Multi-factor authentication (Face, Touch ID, macOS password, PIN).
- Background application monitoring and protection.
- Face enrollment wizard with quality validation.
- Liveness detection (anti-spoofing).
- Activity logging and native notifications.
- Configurable settings for security, sessions, and privacy.

### 1.4.2 Out of Scope

- Cloud-based authentication or biometric sync.
- Mac App Store distribution (requires sandboxing, which prevents app monitoring).
- Intel Mac support (Apple Neural Engine required).
- System login screen replacement.
- File or folder-level encryption (planned for future versions).

# Chapter 2

## Core Features

### 2.1 AI Face Unlock

The primary authentication method uses real-time facial recognition processed entirely on-device.

#### 2.1.1 Technical Details

- **Camera Input:** Live feed captured via `AVFoundation` using `AVCaptureSession` with a front-facing camera (`.builtInWideAngleCamera`, position `.front`).
- **Face Detection:** Apple’s Vision Framework (`VNDetectFaceRectanglesRequest`) provides hardware-accelerated face bounding box detection in under 100 ms.
- **Landmark Detection:** `VNDetectFaceLandmarksRequest` extracts 76 facial landmarks (eyes, nose, mouth, jawline, eyebrows) for alignment.
- **Face Alignment:** An affine transformation is computed from eye positions to normalize the face to a frontal 112×112 pixel image.
- **Feature Extraction:** The aligned face is passed through the ArcFace model (converted to Core ML), which outputs a 512-dimensional embedding vector.
- **Comparison:** Cosine similarity is computed between the live embedding and the stored reference embedding. If the similarity exceeds the configurable threshold (default: 0.55), authentication succeeds.

#### 2.1.2 Cosine Similarity

The similarity between two embedding vectors **a** and **b** is calculated as:

$$\text{similarity}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \times \|\mathbf{b}\|} = \frac{\sum_{i=1}^{512} a_i \cdot b_i}{\sqrt{\sum_{i=1}^{512} a_i^2} \times \sqrt{\sum_{i=1}^{512} b_i^2}}$$

This computation is accelerated using Apple’s `Accelerate` framework (`vDSP`) for SIMD vector operations on Apple Silicon.



## 2.2 Application Locker

Users can protect any installed macOS application by selecting it from a list.

### 2.2.1 App Discovery

Installed applications are discovered by scanning the `/Applications` directory and user-specific `~/Applications`. For each application bundle (`.app`), the following metadata is extracted:

- Display name (from `CFBundleName` in `Info.plist`)
- Bundle identifier (e.g., `com.apple.Safari`)
- Application icon (via `NSWorkspace.shared.icon(forFile:)`)
- File path

### 2.2.2 Data Model

Each protected application is persisted with the following schema:

| Field            | Type   | Description                      |
|------------------|--------|----------------------------------|
| id               | UUID   | Unique identifier                |
| name             | String | Display name                     |
| bundleIdentifier | String | macOS bundle ID                  |
| path             | String | Path to <code>.app</code> bundle |
| isProtected      | Bool   | Whether currently protected      |
| dateAdded        | Date   | When app was added to list       |

Table 2.1: ProtectedApp data model

## 2.3 Background Protection

The application continuously monitors running processes using macOS system APIs.

### 2.3.1 Monitoring Mechanism

Two complementary methods are used for reliability:

1. **NSWorkspace Notifications:** The app subscribes to `NSWorkspace.didLaunchApplicationNot` via `NSWorkspace.shared.notificationCenter`. When any application launches, the notification provides an `NSRunningApplication` object containing the bundle identifier.
2. **Key-Value Observing (KVO):** As a backup, the app observes `NSWorkspace.shared.runningAp` using KVO (`observe(\.runningApplications)`) to detect applications that may not trigger the standard notification (e.g., background-only apps).

## 2.3.2 Interception Flow

When a protected application is detected:

1. The bundle identifier is checked against the protected apps list.
2. If the user has an active authenticated session for that app, access is granted immediately.
3. Otherwise, the protected application is intercepted. The application's windows are hidden using `NSRunningApplication.hide()`, and a `SIGSTOP` UNIX signal is immediately sent to its Process ID (PID) via `kill(pid, SIGSTOP)`. This completely freezes the application in memory, preventing any background processing, network requests, or interactions.
4. A floating authentication panel (`NSPanel` with `.floating` window level) is presented.
5. Upon successful authentication, a `SIGCONT` signal is sent to the PID to unfreeze the process, and the app is unhidden via `NSRunningApplication.unhide()`.
6. If authentication fails, the application remains safely frozen and hidden. No unsaved user data is lost.

## 2.3.3 macOS API Constraints

- The application utilizes POSIX signals to suspend and resume external processes, which provides true OS-level locking.
- The application must be **non-sandboxed** to control other processes. Sandboxed apps are prohibited from modifying external process states.
- This means the application **cannot be distributed via the Mac App Store**.
- Distribution is via notarized DMG, direct download, or Homebrew cask.

## 2.4 Face Enrollment Wizard

The enrollment wizard guides the user through capturing their facial data for future authentication.

### 2.4.1 Process

1. **Pre-checks:** Camera permission is verified. Ambient lighting is assessed using the camera's average luminance. The user is prompted to ensure good lighting and remove obstructions (sunglasses, masks).
2. **Capture Phase:** 20–30 face images are captured at different angles:
  - Center (frontal)
  - Slight left, slight right
  - Slight up, slight down
  - Tilted left, tilted right

Each captured frame undergoes quality validation:

- Face detected within bounding box ( $> 70\%$  of frame)
- Face capture quality score  $> 0.5$  (via `VNDetectFaceCaptureQualityRequest`)

- Both eyes visible (via landmark detection)
  - Sufficient image sharpness (Laplacian variance check)
3. **Embedding Generation:** Each valid capture is aligned and passed through the ArcFace model to produce a 512-dimensional embedding vector.
  4. **Averaging:** All embeddings are element-wise averaged to produce a single reference embedding:

$$\mathbf{e}_{\text{ref}} = \frac{1}{N} \sum_{i=1}^N \mathbf{e}_i$$

where  $N$  is the number of valid captures and  $\mathbf{e}_i$  is the embedding for capture  $i$ .

5. **Normalization:** The averaged embedding is L2-normalized:

$$\hat{\mathbf{e}}_{\text{ref}} = \frac{\mathbf{e}_{\text{ref}}}{\|\mathbf{e}_{\text{ref}}\|}$$

6. **Encryption & Storage:** The normalized embedding is serialized, encrypted, and stored in the macOS Keychain under a dedicated key.
7. **Cleanup:** All raw image data and intermediate embeddings are zeroed out and deallocated from memory. No images are ever written to disk.

## 2.5 Liveness Detection

Liveness detection prevents spoofing attacks where an adversary presents a photo or video of the enrolled user.

### 2.5.1 Tier 1: Vision Framework (No Additional Model)

The primary liveness detection uses Apple’s Vision Framework landmarks:

- **Blink Detection:** The Eye Aspect Ratio (EAR) is computed from eye landmarks. EAR drops below a threshold ( $\sim 0.2$ ) during a blink. Two consecutive blinks within 3 seconds confirms liveness.

$$\text{EAR} = \frac{|p_2 - p_6| + |p_3 - p_5|}{2 \times |p_1 - p_4|}$$

where  $p_1 \dots p_6$  are the six landmarks defining each eye contour.

- **Head Pose Estimation:** Yaw, pitch, and roll angles are computed from facial landmarks. The system issues random challenges:
  - “Look left” (yaw  $< -15$ )
  - “Look right” (yaw  $> 15$ )
  - “Look up” (pitch  $> 10$ )
  - “Nod down” (pitch  $< -10$ )
- **Multi-Frame Consistency:** The system verifies that face landmarks change naturally across multiple consecutive frames rather than remaining static (as with a photograph).

## 2.5.2 Tier 2: MiniFASNet (Optional Advanced Model)

An optional anti-spoofing classification model:

- **Model:** MiniFASNet from the Silent-Face-Anti-Spoofing project.
- **Input:** 80×80 pixel face crop (BGR, normalized to [0, 1]).
- **Output:** Binary classification — real face vs. spoof (photo/screen replay).
- **Conversion:** PyTorch → TorchScript → Core ML via `coremltools`.
- **Usage:** Runs in parallel with ArcFace; both must pass for authentication to succeed.

## 2.6 Multi-Factor Authentication System

FaceLock Pro implements a cascading authentication system with four methods.

### 2.6.1 Authentication Methods

| Method          | Description                | Framework           | Data Storage         |
|-----------------|----------------------------|---------------------|----------------------|
| Face Unlock     | Custom ArcFace recognition | Vision + Core ML    | Keychain (encrypted) |
| Touch ID        | Native fingerprint sensor  | LocalAuthentication | Secure Enclave       |
| macOS Password  | System login password      | LocalAuthentication | Managed by macOS     |
| Application PIN | 4–8 digit user PIN         | Custom              | Keychain (hashed)    |

Table 2.2: Supported authentication methods

### 2.6.2 Authentication Cascade

When a protected application is launched, the following cascade is executed:

1. Face Unlock is attempted automatically (camera activates).
2. If Face Unlock succeeds → access is granted immediately.
3. If Face Unlock fails or times out (5 seconds) → Touch ID is offered.
4. If Touch ID is unavailable or cancelled → macOS password dialog is presented.
5. If the user has enabled a PIN → PIN entry is offered as a final fallback.
6. If all methods fail or are cancelled → access is denied; the app remains locked.

### 2.6.3 Touch ID and macOS Password Integration

Both Touch ID and macOS password authentication are implemented using Apple’s LocalAuthentication framework:

Listing 2.1: Touch ID / macOS Password via LAContext

```
import LocalAuthentication

let context = LAContext()
var error: NSError?

// Check biometric availability
if context.canEvaluatePolicy(
    .deviceOwnerAuthenticationWithBiometrics, error: &error) {

    context.evaluatePolicy(
        .deviceOwnerAuthenticationWithBiometrics,
        localizedReason: "Unlock protected application"
    ) { success, error in
        // Handle result
    }
}
```

The application never accesses, stores, or transmits fingerprint data. All biometric verification is handled by the Secure Enclave through macOS system APIs.

### 2.6.4 PIN Authentication

- PIN length: 4–8 digits, configurable by the user.
- Storage: The PIN is salted, hashed (SHA-256), and stored in the macOS Keychain.
- Retry limits: Configurable maximum attempts (default: 5).
- Cooldown: After max failures, a 60-second cooldown period is enforced.
- The raw PIN is never stored; only the salted hash is persisted.

## 2.7 Session Management

To avoid requiring authentication on every interaction, FaceLock Pro implements robust session management.

### 2.7.1 Per-App Session Tracking

Sessions are tracked individually for each protected application bundle identifier using a timestamp dictionary (`[String: Date]`). This ensures that unlocking one application (e.g., Safari) does not inadvertently unlock another (e.g., Mail).

Additionally, the system implements a **3-second grace period** following a successful authentication. When an app is unlocked and programmatically brought to the foreground via `app.activate()`, macOS fires a subsequent `didActivate` notification. The grace period instructs the monitor to ignore activation events within 3 seconds of a successful authentication, completely eliminating infinite authentication loops without compromising security.

## 2.7.2 Session Timeout Options

| Option              | Duration                              |
|---------------------|---------------------------------------|
| Always Authenticate | 0 seconds (every access)              |
| 5 Minutes           | 300 seconds                           |
| 10 Minutes          | 600 seconds                           |
| 30 Minutes          | 1,800 seconds                         |
| 1 Hour              | 3,600 seconds                         |
| Until Logout        | Indefinite (until macOS session ends) |

Table 2.3: Configurable session timeout durations

## 2.7.3 Auto-Lock Triggers

Sessions are invalidated and re-authentication is required when:

- The Mac enters sleep (`NSWorkspace.willSleepNotification`).
- The screen saver activates (`NSWorkspace.screensDidSleepNotification`).
- The session timeout expires.
- The user manually locks via the menu bar (future feature).

## 2.8 Activity Logging

Every authentication event is recorded for audit purposes.

### 2.8.1 Event Types

| Event                | Icon                               | Description                        |
|----------------------|------------------------------------|------------------------------------|
| App Opened           | arrow.up.forward.app               | A protected app was launched       |
| Auth Success         | checkmark.shield.fill              | Authentication succeeded           |
| Auth Failure         | xmark.shield.fill                  | Authentication failed              |
| App Protected        | lock.fill                          | An app was added to protected list |
| App Unprotected      | lock.open.fill                     | An app was removed from list       |
| Enrollment Completed | person.crop.circle.badge.checkmark | Face data enrolled                 |
| Settings Changed     | gearshape.fill                     | Security settings modified         |
| Session Expired      | clock.badge.xmark                  | Trusted session timed out          |
| Lockout Triggered    | exclamationmark.lock.fill          | Max failed attempts exceeded       |

Table 2.4: Activity log event types

### 2.8.2 Log Entry Schema

Each log entry stores:

| Field               | Type        | Description                             |
|---------------------|-------------|-----------------------------------------|
| id                  | UUID        | Unique identifier                       |
| timestamp           | Date        | Event timestamp                         |
| eventType           | EventType   | Category of event                       |
| appBundleIdentifier | String?     | Target app’s bundle ID                  |
| appName             | String?     | Target app’s display name               |
| authMethod          | AuthMethod? | Method used (Face/TouchID/Password/PIN) |
| wasSuccessful       | Bool        | Whether the action succeeded            |
| confidenceScore     | Float?      | Face recognition confidence (0.0–1.0)   |
| details             | String?     | Additional context or error message     |

Table 2.5: ActivityLogEntry data model

## 2.9 Notifications

The application uses macOS native notifications via the `UserNotifications` framework (`UNUserNotificationCenter`):

- Authentication succeeded for a protected app.
- Authentication failed (with attempt count).
- New application added to the protected list.
- Lockout triggered after maximum failed attempts.
- Security settings were modified.

## 2.10 Settings and Configuration

The application provides comprehensive user-configurable settings organized into four categories:

| Category | Settings                                                                                                           | Storage                 |
|----------|--------------------------------------------------------------------------------------------------------------------|-------------------------|
| General  | Launch at startup, start minimized, theme (system/light/dark)                                                      | UserDefaults            |
| Security | Auth methods toggle, confidence threshold (0.3–0.9), max failed attempts (3–10), lock after sleep, session timeout | UserDefaults            |
| Camera   | Camera device selection, liveness detection toggle                                                                 | UserDefaults            |
| Privacy  | Clear activity logs, delete face data, export/import settings                                                      | UserDefaults + Keychain |

Table 2.6: Settings categories and storage

# Chapter 3

## Technical Architecture

### 3.1 Design Pattern: MVVM

The application follows the Model-View-ViewModel (MVVM) architecture pattern:

- **Model:** Plain Swift structs/classes representing data (e.g., `ProtectedApp`, `FaceEmbedding`, `ActivityLogEntry`). Models are `Codable` for persistence and `Identifiable` for SwiftUI list rendering.
- **View:** SwiftUI views responsible only for rendering UI and forwarding user actions to ViewModels. Views use `@EnvironmentObject`, `@StateObject`, and `@Binding` for state management.
- **ViewModel:** `ObservableObject` classes that encapsulate business logic, transform data for display, and coordinate between services/managers. ViewModels publish state changes via `@Published` properties.

### 3.2 Layered Architecture

The application is divided into five distinct layers, each with clear responsibilities:

#### 3.2.1 Layer 1: Presentation Layer (SwiftUI + AppKit)

- All user interface views built with SwiftUI.
- `NavigationSplitView` for main sidebar navigation.
- `NSPanel` (AppKit) for the floating unlock dialog.
- `NSApplicationDelegateAdaptor` bridges SwiftUI and AppKit lifecycles.
- Screens: Welcome, Dashboard, Protected Apps, Enrollment, History, Settings, Unlock, About.



### 3.2.2 Layer 2: Business Logic Layer (Managers)

| Manager                | Responsibility                                                             |
|------------------------|----------------------------------------------------------------------------|
| AuthenticationManager  | Orchestrates the auth cascade (Face → TouchID → Password → PIN)            |
| FaceRecognitionManager | Manages ArcFace inference, embedding comparison, threshold logic           |
| LivenessManager        | Generates challenges, tracks completion, validates liveness                |
| AppMonitorService      | Subscribes to NSWorkspace notifications, intercepts protected app launches |
| SessionManager         | Tracks per-app sessions, handles timeout and auto-lock                     |
| NotificationManager    | Sends native macOS notifications for auth events                           |
| SettingsManager        | Reads/writes user preferences to UserDefaults                              |
| ActivityLogManager     | Records events to SQLite/SwiftData database                                |

Table 3.1: Business logic managers

### 3.2.3 Layer 3: Machine Learning Layer

| Component          | Responsibility                           | Framework            |
|--------------------|------------------------------------------|----------------------|
| FaceDetector       | Bounding box + landmark detection        | Vision Framework     |
| FaceAligner        | Affine alignment to 112×112              | Core Graphics        |
| EmbeddingGenerator | ArcFace inference → 512-dim vector       | Core ML              |
| CosineSimilarity   | Vector comparison using SIMD             | Accelerate (vDSP)    |
| LivenessClassifier | Real vs. spoof classification (optional) | Core ML (MiniFASNet) |

Table 3.2: Machine learning components

### 3.2.4 Layer 4: Storage Layer

| Store        | Data                                               | Technology         |
|--------------|----------------------------------------------------|--------------------|
| Keychain     | Face embeddings (encrypted), PIN (hashed), secrets | Security Framework |
| Database     | Activity logs, protected apps list                 | SQLite / SwiftData |
| UserDefaults | User preferences, settings                         | Foundation         |

Table 3.3: Storage layer components

### 3.2.5 Layer 5: System Layer

| Framework           | Usage                                                              |
|---------------------|--------------------------------------------------------------------|
| NSWorkspace         | App launch monitoring, sleep/wake notifications, running apps list |
| AVFoundation        | Camera capture session, live video preview, frame extraction       |
| LocalAuthentication | Touch ID and macOS password verification via LAContext             |
| UserNotifications   | Native notification delivery (UNUserNotificationCenter)            |

Table 3.4: System layer frameworks

## 3.3 Project File Structure

The Xcode project is generated from a `project.yml` specification using XcodeGen:

Listing 3.1: Project directory structure

```

AppLockPro/
  App/
    AppLockProApp.swift           // @main SwiftUI entry point
    AppDelegate.swift             // NSApplicationDelegate
    AppState.swift                // Shared observable state
    Info.plist                    // Camera usage, app metadata
    AppLockPro.entitlements        // Camera entitlement
  Models/
    ProtectedApp.swift            // Protected + installed app
    FaceEmbedding.swift           // 512-dim embedding container
    AuthMethod.swift              // Auth method enum
    AuthenticationResult.swift    // Auth attempt result
    ActivityLogEntry.swift        // Log entry + event types
    AppSettings.swift             // Settings + session timeout
  Views/
    ContentView.swift             // Root router
    Components/SidebarView.swift
    Welcome/WelcomeView.swift
    Dashboard/DashboardView.swift
    ProtectedApps/ProtectedAppsView.swift
    Enrollment/EnrollmentWizardView.swift
    History/ActivityHistoryView.swift
    Settings/SettingsView.swift
    Unlock/UnlockDialogView.swift
    About/AboutView.swift
  ViewModels/                    // (Milestone 2+)
  Managers/                      // (Milestone 4+)
  Services/                      // (Milestone 2+)
  AI/                            // (Milestone 3+)
  Utilities/
    Constants.swift               // App-wide constants
    Logger.swift                  // os.Logger categories

```

```
Resources/  
Assets.xcassets           // AccentColor, AppIcon
```

# Chapter 4

## AI Pipeline

### 4.1 Pipeline Overview

The complete AI authentication pipeline executes the following stages sequentially:

1. **Camera Capture** — AVFoundation provides a `CMSampleBuffer` from the front camera at 30 FPS.
2. **Camera Warm-up Delay** — The pipeline explicitly drops the first 10 frames to allow the hardware sensor to auto-adjust exposure and white balance, preventing corrupted embeddings from underexposed frames.
3. **Face Detection** — Vision Framework’s `VNDetectFaceRectanglesRequest` locates faces in the frame.
4. **Landmark Extraction** — `VNDetectFaceLandmarksRequest` identifies 76 facial landmarks.
5. **Quality Assessment** — `VNDetectFaceCaptureQualityRequest` scores the face quality. If the score is below 0.5, the user is prompted to improve lighting.
6. **Face Alignment** — An affine transform normalizes the face using eye positions to a 112×112 image.
7. **Liveness Check** — Eye aspect ratio and head pose are analyzed across multiple frames.
8. **Feature Extraction** — The aligned face is fed to ArcFace (Core ML), producing a 512-dim vector.
9. **L2 Normalization** — The embedding is normalized to unit length.
10. **Cosine Similarity** — The normalized embedding is compared against the stored reference.
11. **Decision** — If similarity  $\geq$  threshold (default 0.55), authentication succeeds.

## 4.2 ArcFace Model

### 4.2.1 Model Specifications

| Property           | Value                           |
|--------------------|---------------------------------|
| Architecture       | ResNet-100 (ArcFace loss)       |
| Source             | InsightFace (open source)       |
| Input Size         | $112 \times 112 \times 3$ (RGB) |
| Output Size        | 512-dimensional float vector    |
| Original Format    | PyTorch (.pth) / ONNX           |
| Target Format      | Core ML (.mlpackage)            |
| Inference Hardware | Apple Neural Engine (ANE)       |
| Expected Latency   | < 200 ms on Apple Silicon       |

Table 4.1: ArcFace model specifications

### 4.2.2 Model Conversion Pipeline

The conversion from PyTorch to Core ML follows this workflow:

1. Download pre-trained ArcFace-R100 weights from the InsightFace model zoo.
2. Load the PyTorch model and set to evaluation mode.
3. Create a dummy input tensor of shape (1, 3, 112, 112).
4. Trace the model using `torch.jit.trace()`.
5. Convert to Core ML using `coremltools.convert()` with image input type.
6. Verify output shape is (1, 512).
7. Save as `.mlpackage` and add to the Xcode project.

Listing 4.1: ArcFace model conversion script

```
import torch
import coremltools as ct

# Load pretrained model
model = load_arcface_r100("arcface_r100.pth")
model.eval()

# Trace with dummy input
dummy = torch.randn(1, 3, 112, 112)
traced = torch.jit.trace(model, dummy)

# Convert to Core ML
mlmodel = ct.convert(
    traced,
    inputs=[ct.ImageType(
        name="faceImage",
        shape=(1, 3, 112, 112),
        scale=1.0/127.5,
```

```

        bias=[-1, -1, -1]
    )],
    minimum_deployment_target=ct.target.macOS14
)
mlmodel.save("ArcFace.mlpackage")

```

### 4.2.3 Preprocessing Requirements

Before passing a face image to the ArcFace model:

1. Detect face bounding box using Vision Framework.
2. Extract eye landmark positions.
3. Compute affine transformation to align eyes horizontally.
4. Apply the transform and crop to  $112 \times 112$  pixels.
5. Convert color space to RGB.
6. Normalize pixel values to  $[-1, 1]$  (scale by  $1/127.5$ , bias  $-1$ ).
7. Convert to `CVPixelBuffer` for Core ML input.

## 4.3 MiniFASNet Anti-Spoofing Model

| Property        | Value                                                |
|-----------------|------------------------------------------------------|
| Architecture    | MiniFASNet (compact CNN)                             |
| Source          | Silent-Face-Anti-Spoofing (GitHub)                   |
| Input Size      | $80 \times 80 \times 3$ (BGR)                        |
| Output          | 2-class (real / spoof)                               |
| Preprocessing   | Face crop with $2.7 \times$ margin, normalize by 255 |
| Original Format | PyTorch (.pth)                                       |
| Target Format   | Core ML (.mlpackage)                                 |

Table 4.2: MiniFASNet model specifications

## 4.4 Hardware Acceleration

All Core ML models are automatically dispatched to the most efficient compute unit by the Core ML runtime:

- **Apple Neural Engine (ANE):** Primary target for both ArcFace and MiniFASNet. Provides the fastest inference with lowest power consumption.
- **GPU (Metal):** Fallback when ANE is unavailable or when the model contains unsupported operations.
- **CPU:** Final fallback; slowest but always available.

The `MLModelConfiguration` is set to `.all` compute units, allowing Core ML to choose the optimal hardware at runtime.

# Chapter 5

## Technology Stack

### 5.1 Complete Technology Table

| Component             | Technology          | Purpose                               |
|-----------------------|---------------------|---------------------------------------|
| Programming Language  | Swift 5.9+          | Type-safe, performant, native         |
| UI Framework          | SwiftUI             | Declarative, modern interface         |
| App Framework         | AppKit              | Window management, system integration |
| Camera                | AVFoundation        | Real-time camera capture              |
| Face Detection        | Vision Framework    | Hardware-accelerated detection        |
| Face Recognition      | ArcFace (Core ML)   | 512-dim embedding extraction          |
| Liveness Detection    | Vision + MiniFASNet | Anti-spoofing                         |
| Hardware Acceleration | Apple Neural Engine | Fast on-device ML inference           |
| Vector Math           | Accelerate (vDSP)   | SIMD cosine similarity                |
| Auth Fallback         | LocalAuthentication | Touch ID + macOS password             |
| Secrets               | Keychain Services   | Encrypted embedding storage           |
| Database              | SQLite / SwiftData  | Activity logs, app list               |
| App Monitoring        | NSWorkspace         | App launch detection                  |
| Notifications         | UserNotifications   | Native macOS alerts                   |
| Logging               | os.Logger           | Unified system logging                |
| Project Generation    | XcodeGen            | .xcodeproj from YAML                  |
| Build System          | xcodebuild          | Command-line builds                   |
| Version Control       | Git                 | Source code management                |

Table 5.1: Complete technology stack

## 5.2 System Requirements

| Requirement       | Specification                               |
|-------------------|---------------------------------------------|
| Operating System  | macOS 14.0 (Sonoma) or later                |
| Processor         | Apple Silicon (M1, M2, M3, M4)              |
| Camera            | Built-in FaceTime HD or compatible external |
| Build Tool        | Xcode 15.0+                                 |
| Project Generator | XcodeGen 2.x                                |
| Disk Space        | ~50 MB (app) + ~100 MB (Core ML models)     |
| RAM               | < 200 MB runtime footprint                  |

Table 5.2: System requirements

## 5.3 Build and Run Instructions

Listing 5.1: Building from the command line

```
# Install XcodeGen (one-time)
brew install xcodegen

# Clone the repository
git clone https://github.com/ThanojBuddhima/applockPro.git
cd applockPro

# Generate the Xcode project from project.yml
xcodegen generate

# Build the application
xcodebuild build \
  -project AppLockPro.xcodeproj \
  -scheme AppLockPro \
  -configuration Debug \
  -destination 'platform=macOS'

# Run the built app
open ~/Library/Developer/Xcode/DerivedData/AppLockPro-*/\
  Build/Products/Debug/FaceLock\ Pro.app
```

Alternatively, open `AppLockPro.xcodeproj` in Xcode and press `Cmd+R`.



# Chapter 6

## Security Design

### 6.1 Security Principles

1. **Zero Cloud Upload:** All biometric data, images, and embeddings remain on the local device. No network requests are made for authentication.
2. **No Image Persistence:** Raw face images captured during enrollment or authentication are never written to disk. They exist only in memory and are deallocated immediately after embedding generation.
3. **Keychain-Only Secrets:** Face embeddings and PINs are stored exclusively in the macOS Keychain, which provides hardware-backed AES-256 encryption via the Secure Enclave.
4. **Local Inference:** All AI models run on-device via Core ML, utilizing the Apple Neural Engine.
5. **Hardened Runtime:** The application is built with Hardened Runtime enabled, which restricts code injection, memory access, and dynamic library loading.
6. **Principle of Least Privilege:** The application only requests the permissions it needs (camera access, accessibility for app monitoring).

### 6.2 Data Protection Matrix

| Data Type           | Storage      | Encryption           | Lifecycle                    |
|---------------------|--------------|----------------------|------------------------------|
| Face embeddings     | Keychain     | AES-256 (hardware)   | Persisted until user deletes |
| Application PIN     | Keychain     | SHA-256 hash + salt  | Persisted until user changes |
| Activity logs       | SQLite       | Application-level    | Clearable by user            |
| Protected apps list | SQLite       | Application-level    | User-managed                 |
| User preferences    | UserDefaults | None (non-sensitive) | Persisted                    |
| Raw face images     | RAM only     | N/A                  | Deleted immediately          |
| Camera frames       | RAM only     | N/A                  | Not retained                 |

Table 6.1: Data protection matrix

## 6.3 Threat Model

| Threat            | Attack Vector                                             | Mitigation                                                                             |
|-------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------|
| Photo replay      | Attacker holds a photo of the user in front of the camera | Liveness detection: blink detection, head movement challenges, multi-frame consistency |
| Video replay      | Attacker plays a video of the user on a screen            | MiniFASNet anti-spoofing classifier, random challenge sequence                         |
| Embedding theft   | Attacker extracts stored face embeddings from disk        | Keychain AES-256 encryption, Hardened Runtime prevents memory inspection               |
| Brute-force PIN   | Attacker repeatedly guesses the PIN                       | Max attempts (default 5), cooldown period (60s), temporary lockout                     |
| App bypass        | User unhides protected app via Mission Control            | Documented limitation; future mitigation via Endpoint Security Framework               |
| Process injection | Attacker injects code to bypass authentication checks     | Hardened Runtime, code signing, non-sandboxed but signed with Developer ID             |

Table 6.2: Threat model and mitigations

## 6.4 Keychain Integration

The macOS Keychain is used for all sensitive data storage:

Listing 6.1: Keychain storage example

```
import Security

func storeEmbedding(_ data: Data, forKey key: String) -> Bool {
    let query: [String: Any] = [
        kSecClass as String: kSecClassGenericPassword,
        kSecAttrAccount as String: key,
        kSecValueData as String: data,
        kSecAttrAccessible as String:
            kSecAttrAccessibleWhenUnlockedThisDeviceOnly
    ]

    // Delete existing item if present
    SecItemDelete(query as CFDictionary)

    // Add new item
    let status = SecItemAdd(query as CFDictionary, nil)
    return status == errSecSuccess
}
```

The `kSecAttrAccessibleWhenUnlockedThisDeviceOnly` flag ensures embeddings are only accessible when the device is unlocked and cannot be transferred to another device via backup.

# Chapter 7

## User Interface Design

### 7.1 Design Principles

The interface follows Apple’s macOS Human Interface Guidelines:

- Native macOS look and feel using SwiftUI.
- Support for Light Mode and Dark Mode.
- Smooth animations (spring, easeInOut).
- SF Symbols for consistent iconography.
- Keyboard shortcut support.
- Accessibility labels for VoiceOver compatibility.
- Retina display optimization.

### 7.2 Application Screens

#### 7.2.1 Welcome / Onboarding (4 Steps)

A guided wizard presented on first launch:

1. **Welcome:** Introduction with feature highlights (app protection, face unlock, offline privacy).
2. **Permissions:** Camera and accessibility permission requests with status indicators.
3. **Enrollment Intro:** Explanation of the face capture process.
4. **Complete:** Confirmation that setup is finished; “Get Started” button.

Progress is shown via animated capsule indicators at the top.

#### 7.2.2 Dashboard

The main screen after onboarding, featuring:

- **Status Badge:** Shows enrollment status (“Face Enrolled” / “Not Enrolled”).
- **Stat Cards:** Protected apps count, successful unlocks today, failed attempts today, session status.
- **Quick Actions:** Add app, re-enroll face, view history, settings.
- **Recent Activity:** Last 5 authentication events.

### 7.2.3 Protected Apps

- Search bar to filter protected apps.
- List of protected apps with name, bundle ID, icon, and toggle switch.
- “Add App” button opens a sheet with all installed applications.
- Swipe-to-delete for removing apps from protection.

### 7.2.4 Face Enrollment Wizard

Five-state wizard:

1. **Ready:** Instructions and lighting requirements.
2. **Capturing:** Live camera preview with angle guidance and progress bar.
3. **Processing:** Spinner while embeddings are generated and encrypted.
4. **Complete:** Success confirmation.
5. **Error:** Failure message with “Try Again” option.

### 7.2.5 Unlock Dialog

A floating panel (`NSPanel`) displayed above all windows:

- Shows the requesting application’s name and icon.
- Camera preview for Face Unlock (primary method).
- Method switcher: buttons to switch between Face, Touch ID, Password, and PIN.
- Authentication status feedback (authenticating, success, failure, locked out).

### 7.2.6 Activity History

- Filter chips for each event type (All, Auth Success, Auth Failure, etc.).
- Chronological list with event icon, name, app name, confidence score, and relative timestamp.
- “Clear All” button with confirmation dialog.

### 7.2.7 Settings

Four-tab interface:

- **General:** Launch at startup, start minimized, theme (System/Light/Dark).
- **Security:** Auth method toggles, confidence threshold slider (30%–90%), max attempts stepper, lock after sleep, session timeout picker.
- **Camera:** Camera device picker, liveness detection toggle.
- **Privacy:** Clear logs, delete face data (with confirmation dialogs), export/import settings.

### 7.2.8 About

Application information: icon, name, version, description, technology badges, and copyright.

## 7.3 Navigation Architecture

The application uses `NavigationSplitView` with a sidebar containing three sections:

- **Main:** Dashboard, Protected Apps
- **Security:** Face Enrollment, Activity History
- **Application:** Settings, About

Navigation selection is tracked via the shared `AppState` object, which is injected into the view hierarchy using `@EnvironmentObject`.

# Chapter 8

## Performance

### 8.1 Performance Targets

| Metric            | Target   | Method                    | Hardware            |
|-------------------|----------|---------------------------|---------------------|
| Face detection    | < 100 ms | Vision Framework          | GPU                 |
| Face recognition  | < 200 ms | ArcFace Core ML           | Apple Neural Engine |
| Cosine similarity | < 1 ms   | Accelerate vDSP           | CPU (SIMD)          |
| Total unlock time | < 500 ms | End-to-end pipeline       | ANE + GPU + CPU     |
| CPU usage (idle)  | < 10%    | Event-driven monitoring   | CPU                 |
| Memory footprint  | < 200 MB | Model + UI + buffers      | RAM                 |
| Battery impact    | Minimal  | ANE offloading            | Neural Engine       |
| App monitoring    | < 1 ms   | NSWorkspace notifications | CPU (event)         |

Table 8.1: Performance targets with hardware mapping

### 8.2 Optimization Strategies

- **Event-Driven Monitoring:** App monitoring uses NSWorkspace notifications rather than polling, consuming near-zero CPU when no protected app is launched.
- **Lazy Model Loading:** Core ML models are loaded into memory only when face authentication is first requested, not at app startup.
- **Frame Skipping:** During live camera preview, face detection is performed on every 3rd frame (10 FPS) rather than every frame (30 FPS) to reduce CPU/GPU load.
- **Accelerate Framework:** Cosine similarity computation uses `vDSP.dot()` and `vDSP.sumOfSquares()` for SIMD-optimized vector operations.
- **Memory Management:** Camera frame buffers are released immediately after processing. Auto-release pools are used for tight loops.

# Chapter 9

## Implementation Roadmap

Development is divided into seven milestones, each producing a functional increment:

### 9.1 Milestone 1: Project Setup & UI Shell (Complete)

- Xcode project generation via XcodeGen.
- Complete folder structure (App, Models, Views, ViewModels, Managers, Services, AI, Utilities).
- All SwiftUI screens implemented as navigable stubs.
- Data models for all entities.
- Build verification: `xcodebuild` succeeds.

### 9.2 Milestone 2: Camera & Face Detection

- `CameraService`: AVFoundation capture session with front camera.
- Camera preview via `AVCaptureVideoPreviewLayer` wrapped in `NSViewRepresentable`.
- `FaceDetector`: Vision Framework face detection and landmark extraction.
- Real-time face bounding box overlay on camera feed.
- `FaceAligner`: Affine alignment to 112×112 using eye positions.
- Face quality validation (lighting, sharpness, eyes visible).

### 9.3 Milestone 3: AI Face Recognition

- Python script for ArcFace PyTorch → Core ML conversion.
- `EmbeddingGenerator`: Core ML inference wrapper.
- `CosineSimilarity`: Accelerate-powered vector comparison.
- Enrollment pipeline: capture → detect → align → embed → average → encrypt → store.
- Authentication pipeline: capture → detect → align → embed → compare → decision.



## 9.4 Milestone 4: App Monitoring & Protection

- `AppMonitorService`: `NSWorkspace` notifications + KVO.
- Installed apps browser (scan `/Applications`).
- App hide/terminate on unauthorized launch.
- Floating unlock panel (`NSPanel`).
- Session management with configurable timeouts.
- Sleep/screen lock event handling.

## 9.5 Milestone 5: Multi-Factor Authentication

- Touch ID + macOS password via `LAContext`.
- PIN setup, verification, and Keychain storage.
- Authentication cascade logic.
- `KeychainService` wrapper.
- Activity logging to `SQLite/SwiftData`.
- Failed attempt tracking, cooldown, and lockout.
- Native macOS notifications.

## 9.6 Milestone 6: Liveness Detection

- Blink detection via eye aspect ratio.
- Head pose estimation from landmarks.
- Random challenge generator and UI.
- Multi-frame consistency verification.
- Optional MiniFASNet integration.

## 9.7 Milestone 7: Polish & Release

- Smooth animations and transitions.
- App icon design.
- Keyboard shortcuts and accessibility.
- Performance profiling with Instruments.
- Error handling for all edge cases.
- Code signing with Developer ID.
- Notarization for Gatekeeper.
- DMG packaging.
- Documentation (README, architecture doc, user guide).

## 9.8 Estimated Timeline

| Week | Milestone              | Deliverable                             |
|------|------------------------|-----------------------------------------|
| 1    | Setup & Learning       | Project structure, SwiftUI basics       |
| 2    | Milestone 1 (complete) | Full UI skeleton, all screens navigable |
| 3    | Milestone 2            | Live camera + face detection            |
| 4    | Milestone 3            | ArcFace integration, enrollment, auth   |
| 5    | Milestone 4            | App monitoring, protection, unlock flow |
| 6    | Milestone 5            | Touch ID, PIN, logging, notifications   |
| 7    | Milestone 6            | Liveness detection                      |
| 8    | Milestone 7            | Polish, testing, release candidate      |

Table 9.1: Development timeline

# Chapter 10

## Testing Strategy

### 10.1 Unit Tests

- **CosineSimilarity**: Verify correct similarity scores for known vectors, edge cases (zero vectors, identical vectors, orthogonal vectors).
- **EmbeddingGenerator**: Test with mock Core ML model, verify output dimensions.
- **KeychainService**: Store/retrieve/delete operations with test keys.
- **SessionManager**: Timeout expiration, auto-lock triggers, session creation/invalidation.
- **ActivityLogManager**: Event recording, filtering, clearing.
- **AppMonitorService**: Protected app lookup, notification handling.

### 10.2 Integration Tests

- Full enrollment pipeline: 20 captures → embeddings → average → Keychain storage.
- Full authentication pipeline: camera → detection → alignment → embedding → comparison.
- Authentication cascade: Face failure → Touch ID → Password → PIN.
- App monitoring: Launch protected app → detect → block → authenticate → unlock.

### 10.3 Manual Verification Checklist

- Camera preview displays live feed.
- Face detection draws bounding box in real-time.
- Enrollment captures faces and stores encrypted embedding.
- Face unlock succeeds for enrolled user within 500 ms.
- Face unlock rejects unknown faces.
- Protected app is hidden on unauthorized launch.
- Protected app unlocks after successful authentication.
- Touch ID and PIN fallbacks work correctly.
- Auto-lock triggers on sleep.
- Activity log records all events.
- Settings persist across app restarts.

- Launch at login works.
- CPU stays below 10% at idle.
- Memory stays under 200 MB.

# Chapter 11

## Risk Assessment

| Risk                           | Likelihood | Impact | Mitigation                                                                               |
|--------------------------------|------------|--------|------------------------------------------------------------------------------------------|
| ArcFace model conversion fails | Medium     | High   | Fallback: use MobileFaceNet or ONNX Runtime with CoreML Execution Provider               |
| App hiding is bypassable       | High       | Medium | Document limitation; plan Endpoint Security Framework upgrade in v2                      |
| Camera permission denied       | Low        | Medium | Graceful fallback to Touch ID / password only                                            |
| Performance exceeds targets    | Medium     | Medium | Profile early with Instruments; use ANE; reduce model size if needed                     |
| MiniFASNet conversion issues   | Medium     | Low    | Defer to Vision-only liveness (Tier 1) which needs no extra model                        |
| Keychain access issues         | Low        | Low    | Proper entitlements; test on clean macOS install                                         |
| ArcFace licensing conflict     | Low        | High   | Use non-commercial for learning; purchase license or switch model for commercial release |

Table 11.1: Risk assessment matrix

# Chapter 12

## Future Enhancements

Future versions of FaceLock Pro may include:

- **Multiple User Profiles:** Allow multiple users to enroll and authenticate on a shared Mac.
- **Apple Watch Proximity Unlock:** Use Bluetooth proximity with a paired Apple Watch to unlock apps without camera.
- **Geofencing:** Automatically disable authentication at trusted locations (e.g., home, office) using Core Location.
- **Secure File Vault:** Encrypt and protect individual files and folders with face authentication.
- **Endpoint Security Framework:** Intercept process launches at the kernel level for stronger app blocking (requires Apple entitlement).
- **Encrypted Backups:** Export and import encrypted face data and settings for device migration.
- **AI Anomaly Detection:** Detect unusual authentication patterns (e.g., repeated failures from an unknown face).
- **Plugin Architecture:** Allow third-party extensions for custom protection rules.
- **Enterprise Management:** MDM integration for corporate deployment.
- **Menu Bar Controls:** Quick lock/unlock, status, and settings from the macOS menu bar.
- **Browser Extension:** Protect browser tabs and bookmarks within Safari and Chrome.
- **Clipboard Protection:** Clear clipboard contents after a configurable timeout.
- **Remote Lock:** Lock all protected apps remotely via a companion iOS app.

# Chapter 13

## Conclusion

FaceLock Pro addresses a real gap in macOS security by providing application-level biometric protection that is private, fast, and entirely offline. By leveraging Apple Silicon’s Neural Engine, the Vision framework, and the ArcFace facial recognition model, the application achieves sub-500-millisecond authentication without compromising user privacy.

The modular MVVM architecture, phased development approach, and clear separation of concerns ensure that the project remains maintainable as new features are added. The milestone-based roadmap transforms a complex undertaking into a series of achievable increments, each delivering a working product.

This project demonstrates proficiency in Swift, SwiftUI, Core ML, computer vision, biometric security, and native macOS development — making it a comprehensive showcase of modern Apple platform engineering.

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